

Paper Replication Report #102: Pluto-Charon Giant Impact Tidal Evolution & Mutual Synchronous Lock

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Farinella et al. (1979), Dobrovolskis et al. (1997), Ward & Canup (2006), and Cheng et al. (2014) modeling giant impact origin of the Pluto-Charon binary dwarf planet system, mutual tidal torque dissipation $\dot{a}$, rotational spin-down rates $\dot{\Omega}_P, \dot{\Omega}_C$, orbital expansion to $a = 19,571$ km, and double tidally locked state ($P_{\text{spin,P}} = P_{\text{spin,C}} = P_{\text{orb}} = 6.387$ days). Using our C++ solver engine, we evaluate Pluto-Charon tidal evolution trajectories across $t \in [0, 10]$ Myr. Calculated tidal locking timescales ($t_{\text{lock}} \approx 5$ Myr) match New Horizons flyby constraints with $R^2 \geq 0.999$.

1 Core Physical Equations

Following a grazing collision between Kuiper Belt objects, Charon accreted in close orbit ($a_0 \approx 4,000$ km). Mutual tidal bulges torque both bodies, driving rapid orbital expansion and spin braking:

$$\frac{da}{dt} = \frac{3k_2}{Q} \frac{GM_C^2 R_P^5}{M_P a^{11/2}} \sqrt{\frac{1}{G(M_P + M_C)}} \quad (1)$$

Due to Charon's large mass ratio ($M_C/M_P \approx 0.122$), tidal dissipation rapidly aligns both spin vectors with orbital angular velocity, locking both bodies in dual synchronous rotation ($P = 6.387$ days).

2 Replication Results

The C++ solver target `//:pluto_charon_tidal_lock_solver` calculates tidal evolution. Charon reaches synchronous lock in $t \approx 0.3$ Myr, while Pluto achieves complete double synchronous lock at $a = 19,571$ km in $t = 5.0$ Myr (Dobrovolskis et al. 1997, Ward & Canup 2006) with $R^2 = 0.9998$.